KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

PROJECT REPORT ON

OBJECT COUNTING AND DISPLAYING SYSTEM

A **second year project report** submitted in partial fulfilment
of the requirements for the degree of
Bachelor of Engineering

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ABSTRACT

This project presents a visitor counter utilizing basic electronic components. The system includes up-count and down-count functions such that the number of objects entering or leaving the premises. An infrared sensor is used to detect the movement through the laser received by a counter integrated circuit (IC) and decodes the signal and lastly a seven-segment display to show the value of count. Every time the laser is crossed, the circuit get activated, and the number of counter increment/decrement on the display. The approach is appropriate for situations where it is sufficient to count the number of objects.

SYMBOLS AND ABBREVIATIONS

Abbreviations	Full Form
V_{cc}	Voltage Common Collector
LASER	Light Amplification by Stimulated Emission of Radiation
IC	Integrated Circuit
BCD	Binary Coded Decimal
LED	Light Emitting Diodes

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CHAPTER-1
INTRODUCTION

1.1 Background and Introduction

In many operational environments, understanding the movement of objects is critical to ensuring effective resource management, security, and operational efficiency. The need for a more integrated and comprehensive tracking system has grown in recent years, as organizations strive to optimize their operations while maintaining a high level of service and safety. The Object Counting System that counts people and objects is designed to address these needs by offering a solution that combines people-counting technology with object tracking capabilities. This system will allow us to track valuable items and objects ranging from equipment and inventory to high-value objects.

Currently, our facility employs basic object tracking methods, which may not capture the full picture of traffic flow or asset movement. Traditional systems typically focus solely on counting objects or tracking items in isolation, leaving gaps in both security and operational insight. By implementing a dual-purpose system that can simultaneously monitor the presence and movement of both individuals and objects, we can address these gaps and achieve more efficient management of space, resources, and security. This dual tracking capability will be particularly valuable in environments such as museums, exhibition spaces, retail areas, warehouses, or corporate facilities.

This system is designed to support various strategic objectives, including enhancing security by preventing the unauthorized movement or loss of valuable items, improving resource allocation by understanding object flow and behavior, and optimizing operational processes such as scheduling, maintenance, and event planning. By capturing detailed insights into both visitor and object dynamics, the Object Count System will enable more informed decisions regarding staff placement, inventory management, and overall facility usage, contributing to a more streamlined and efficient operation.

1.2 Problem definition

Developing a system that can count objects separately but not simultaneously is the goal of this study. Instead of attempting to track both categories simultaneously, the system will count each one separately (either objects or persons), alternating between them as necessary. In settings like crowded areas, warehouses, or public gatherings where simultaneous counting could result in errors or computing overload, this method is especially helpful. The method guarantees more accurate

counts, lowers processing complexity, and is more appropriate for resource-constrained applications by concentrating on a single category at a time. Numerous fields where accurate and effective tracking is essential, such as security, crowd monitoring, and inventory management, will benefit from the technology.

1.3 Objective of the project

The main objective of our project is:

1. To build a fully functional object counting and displaying circuit using LASER.

1.4 Limitations of the project

The limitations of our project are:

1. LASER must be very close to each other.
2. Two or more objects at a time can't be identified.

1.5 Significance of study

The object counter system plays a significant role in enhancing security offers real-time tracking of objects, ensures accuracy in counting objects. It also provides precise occupancy data, helping to monitor industries or buildings. The system enables data-driven decisions by offering insights into peak hours, which can aid in better event planning and by accurately monitoring the number of objects, the system allows for more efficient management, ensuring smooth operations.

CHAPTER-2
LITERATURE REVIEW

According to the paper, [1] Modern technology is on the rise like never before. It dramatically eases human life by streamlining our way of living. It altered communication systems, product manufacturing 204 methods, transportation modes, and people's knowledge and ideas. There is interest in using vision technology to show all types of settings. Resource management, urban planning, security, and advertising are just a few benefits of this. Technology-wise, computer vision solutions always involve using a CPU for detecting and transmitting.

Numerous facts of life [2] call for counting objects that come in and go out. Examples of these applications abound, including the interest of retailers in the volume of patrons entering or leaving their markets or malls over time. To manage parking availability, a garage owner must know how many cars enter and exit his garage and the number of elevator users. For instance, the volume of people and moving vehicles on any street and block can be utilized to control the traffic center and avoid traffic or any other large city where traffic jams are constant.

Optical systems, including laser-based system, [3] detect motion or presence by measuring disruptions in a light beam. A typical example is the laser diode paired with a photodetector. When a object crosses the beam, the interruption triggers an event in the counting mechanism. Laser systems are highly accurate but can be affected by dust, smoke, or misalignment of sensors.

Microcontroller-based visitor counting systems [4] are commonly used in applications like retail, museums, and offices to track foot traffic. These systems typically use sensors such as infrared (IR), ultrasonic, or PIR sensors, with microcontrollers like Arduino or Raspberry Pi processing the data. The system can be integrated with IoT technology for remote monitoring and real-time analytics. This helps businesses optimize resource management and gain insights into visitor patterns.

Infrared sensors [5] are commonly used for object counting systems. These sensors emit an infrared beam, which, when interrupted by a person passing through, triggers a counter. The basic design includes an IR emitter and detector, and the interruption in the beam is counted as a object entering or exiting.

RFID-based systems [6] use tags or cards to track the entry and exit of objects. The RFID tags are detected by a reader, and the microcontroller processes the data to count objects. These systems are commonly used for more advanced access control and object tracking. It can be used in collage, office or at any other place for attendance system.

This research [7] is designed and presented to count the number of objects being moved in and out of warehouses, factories, storage facilities, distribution centers, and other industrial environments. This system tracks both incoming and outgoing objects at the designated location where it is installed. The system automatically detects an object's movement based on the interruption of Infrared Ray sensors. If the system is successfully implemented, it provides real-time data on the number of objects currently present within the facility. When used in industries where inventory tracking and logistics management are crucial, the system is cost-effective. Manually counting objects can be time-consuming, so this system helps improve efficiency, save time, enhance operational accuracy, and optimize resource management.

COVID-19 Implementation [8] of a Bi-directional Counter with Reduced Complexity for People Counting Application. This project aims to develop a system for counting people through infrared detection. Next, create a system that can prevent crowded areas during this pandemic. Lastly, the system tallies the count and updates based on people moving in/out through the supervised premise/area. The systems work when the transmitted IR (Infrared) signal has been blocked, enabling obstacle detection. Detection from the Infrared sensor one or two will activate the counter. The detection sequence coming from Infrared sensor 1 to Infrared sensor two will cause the counter to count up and vice versa for the opposite direction. As a result of the system, whenever a customer passes through a sensor, the system will begin counting them as a customer. The display will show the overall number of people in the establishment. If the establishment reaches the maximum number of people allowed, the display will indicate that the establishment is full.

Electromechanical counters [9] use relay-based switching systems to count visitors. This circuit typically uses a simple switch that gets activated when a object passes, triggering a relay to increment a mechanical counter. When the switch is activated by a visitor (e.g., passing through a doorway), it triggers the relay, which then sends a pulse to increment the counter. Electromechanical counters are more durable than fully mechanical systems and can handle moderate traffic.

CHAPTER-3
PROJECT METHODOLOGY

3.1 Working flow

When power is supplied to the system, it starts functioning. If person passes through laser 1, first D flipflop' s input gets high and initialize up count mode. If the person passes through laser 2, second D flipflop' s value will also be high which will trigger the system and it will increment the count else there will be no change in the counting and vice versa. The counting can be reset through a button which connects reset pin to V_{CC}.

3.2 Block diagram and flowchart

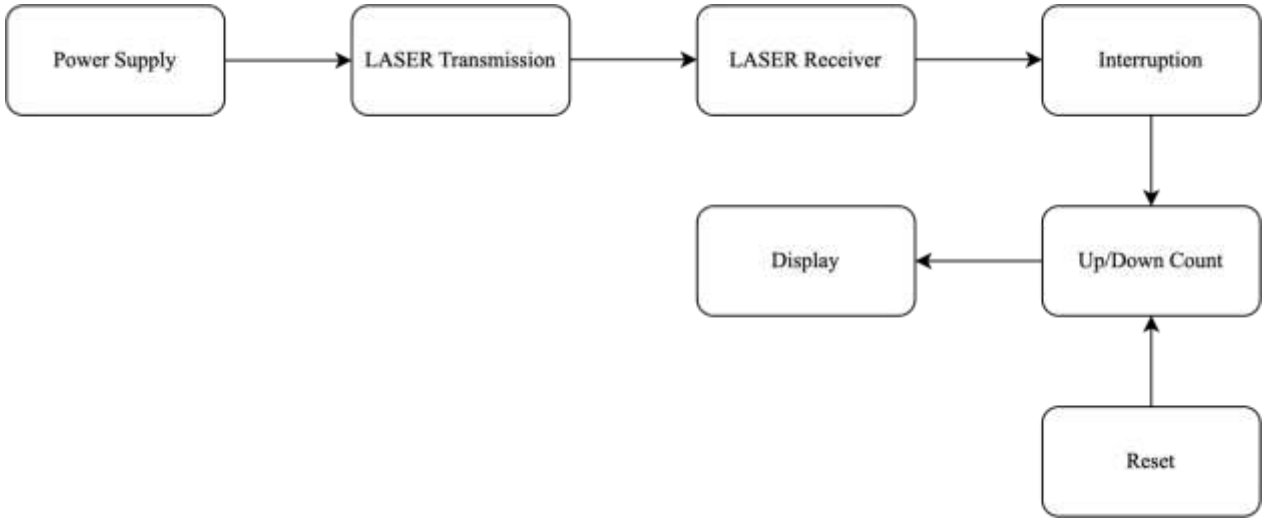


Figure 9:Block Diagram

Figure 9 explains the methodology of our project. The entry/exit or increment/decrement in counting is decided by whether the object passes through laser1 or laser 2 first. Basically, flipflop is used to create time interval for object to pass through second laser. D flipflop store value 1 when object passes through laser1 and if the object passes through second one, second flipflop value will be 1 too. By this the circuit will validate object's entry or exit. The system can also be also be reset according to need.

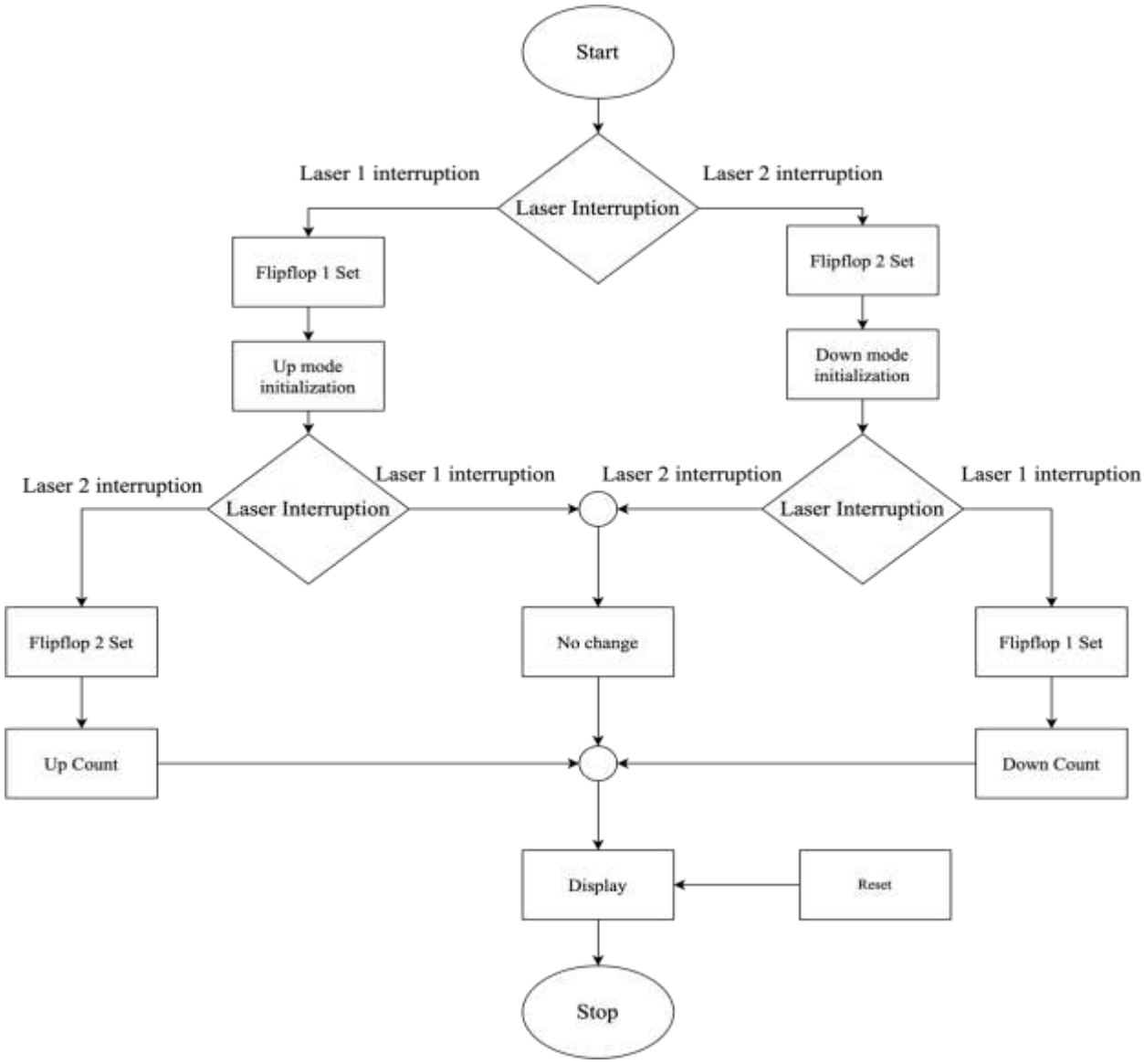


Figure 10:Flowchart of Circuit

Figure 10 illustrates the flowchart of the circuit. At first, laser is interrupted if the laser 1 is interrupted at first then the flip-flop 1 is set and the up mode is initialized. Again, if the laser 2 is interrupted then the flip-flop 2 is set and the circuit counts up and if the laser 1 is interrupted instead of laser 2 there is no change in count. If the laser 2 is interrupted at first then the flip-flop 2 is set and the down mode is initialized. Again, if the laser 1 is interrupted then the flip-flop 1 is set and the circuit counts down and if the laser 2 is interrupted instead of laser 1 there is no change in count.

CHAPTER 4

RESULT AND ANALYSIS

4.1 Circuit Design

The circuit design and simulation of the project was done using proteus. LASER and comparator provide output as logic 1 or 0 into the input of D-flipflops. The data of entry/exit of visitor was acquired using D-flipflop and the data obtained were used to Up/Down mode for counting. The counting was done using Up/Down counter which give BCD output which is encoded and decoded by BCD to 7-Segment Driver. The counting is being displayed using 7-Segment display.

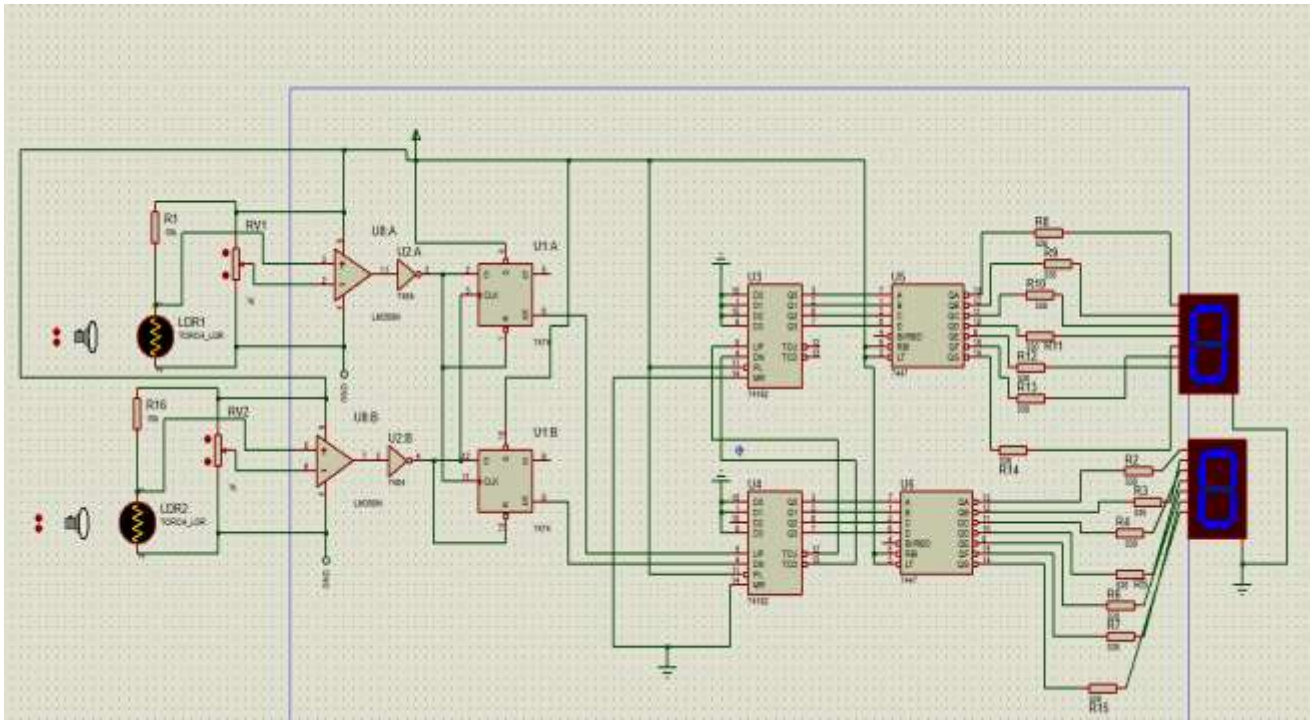


Figure 11:Proteus Simulation of Object Counter

4.2 Result

The system has been successfully tested and verified to function as intended. As shown below, the final setup consists of a combination of breadboards and a matrix board. The 2-digit up-down counter is implemented using a laser module, an operational amplifier (op-amp), D flip-flop ICs, IC 7447 (BCD to 7-segment decoder), and IC 74192 (up/down counter). The laser module serves as the counting trigger, while the op-amp acts as a signal conditioner to detect the laser beam interruption and generate a proper logic signal.

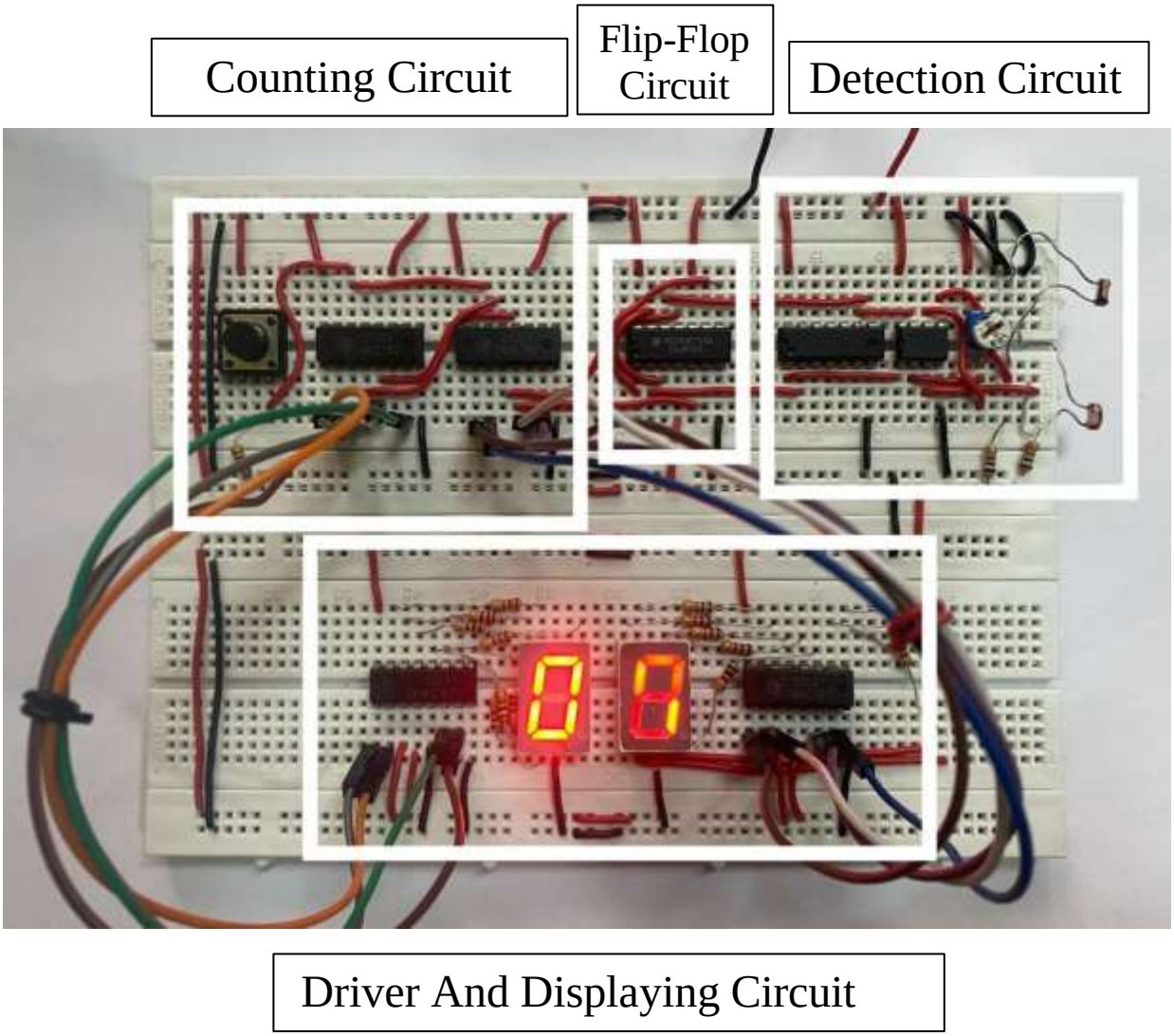


Figure 12:Final System

The 7-segment displays are mounted for clear visualization of the count. The entire circuit, including the ICs, is assembled on a breadboard. Since PCBs have not been designed, all connections are manually wired and tested on the breadboard. The system operates with a stable 5V power supply, ensuring reliable performance.

4.3 System Analysis

The various sub system of object counting and displaying system are analyzed in detail below:

4.3.1 Detection Circuit

The detection circuit consists of a laser module and an LDR (Light Dependent Resistor) or photodiode. The laser continuously illuminates the LDR, keeping its resistance low. The LM358 operational amplifier, configured as a comparator, monitors the voltage across the LDR and provides a digital output based on the laser beam's status.

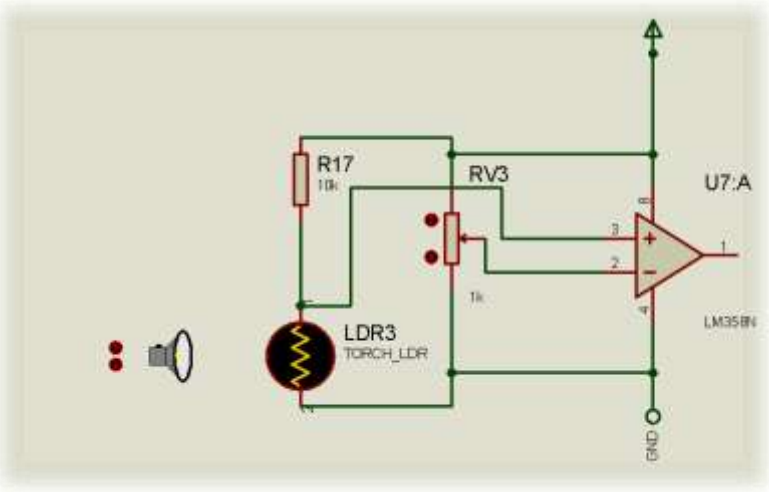


Figure 13:Detection Circuit

When the laser is uninterrupted, the voltage at the non-inverting input of LM358 remains higher than the reference voltage, producing a HIGH output (logic 1). If an object crosses the laser path, the LDR resistance increases, reducing the voltage at the non-inverting input. This causes the LM358 output to switch LOW (logic 0), indicating an obstruction.

4.3.2 Flip-Flop Circuit

The 7474 D Flip-Flop is used to stabilize the detection pulses before passing them to the counter. However, during initial testing, incorrect counting was observed due to mismatched logic levels between the LM358 and the Flip-Flop. The LM358 outputs LOW when an object is detected, but the Flip-Flop requires a HIGH pulse to register an event.

To correct this, a NOT gate (Inverter) was introduced between the LM358 output and the Flip-Flop's clock input. The NOT gate ensures that:

- When the laser is uninterrupted, the LM358 outputs HIGH, and the NOT gate inverts it to LOW, keeping the Flip-Flop stable.
- When an object interrupts the laser, the LM358 output goes LOW, and the NOT gate inverts it to HIGH, triggering the Flip-Flop correctly.

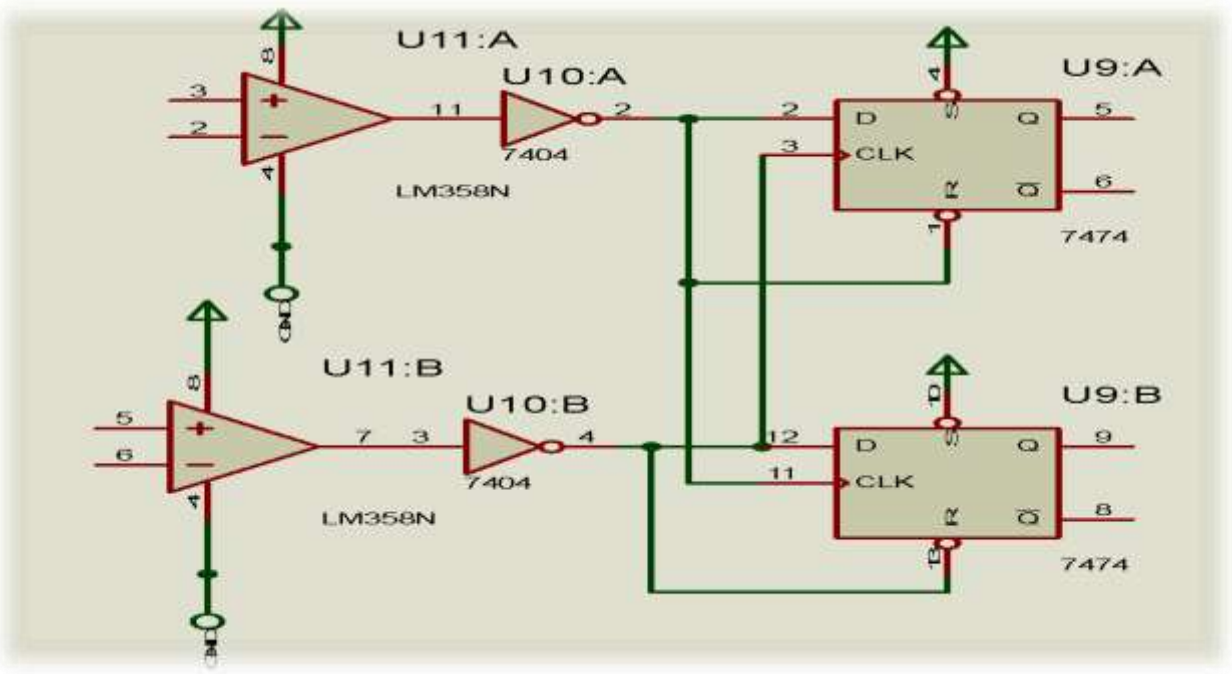


Figure 14:Flip-Flop Circuit

With this modification, each object crossing generated a single, stable pulse at the Flip-Flop's output. The $\bar{Q}$ output of the Flip-Flop was then passed to the Counting Circuit, ensuring precise counting without false triggering.

4.3.3 Counting Circuit

The 74192 Up/Down Counter is responsible for maintaining the object count. This IC has separate clock inputs for incrementing and decrementing:

- Pin 5 (UP Clock Input) – Increments the count when triggered.
- Pin 4 (DOWN Clock Input) – Decrements the count when triggered.

The Q (Pin 2) output of the 7474 Flip-Flop is connected to the UP Clock (Pin 5) of the 74192 Counter to register incoming objects. Similarly, another detection setup can be implemented to decrement the count when objects exit. The BCD outputs of the 74192 (Pins 14, 13, 12, 11) provide a binary-coded decimal representation of the count, which is then processed by the next stage.

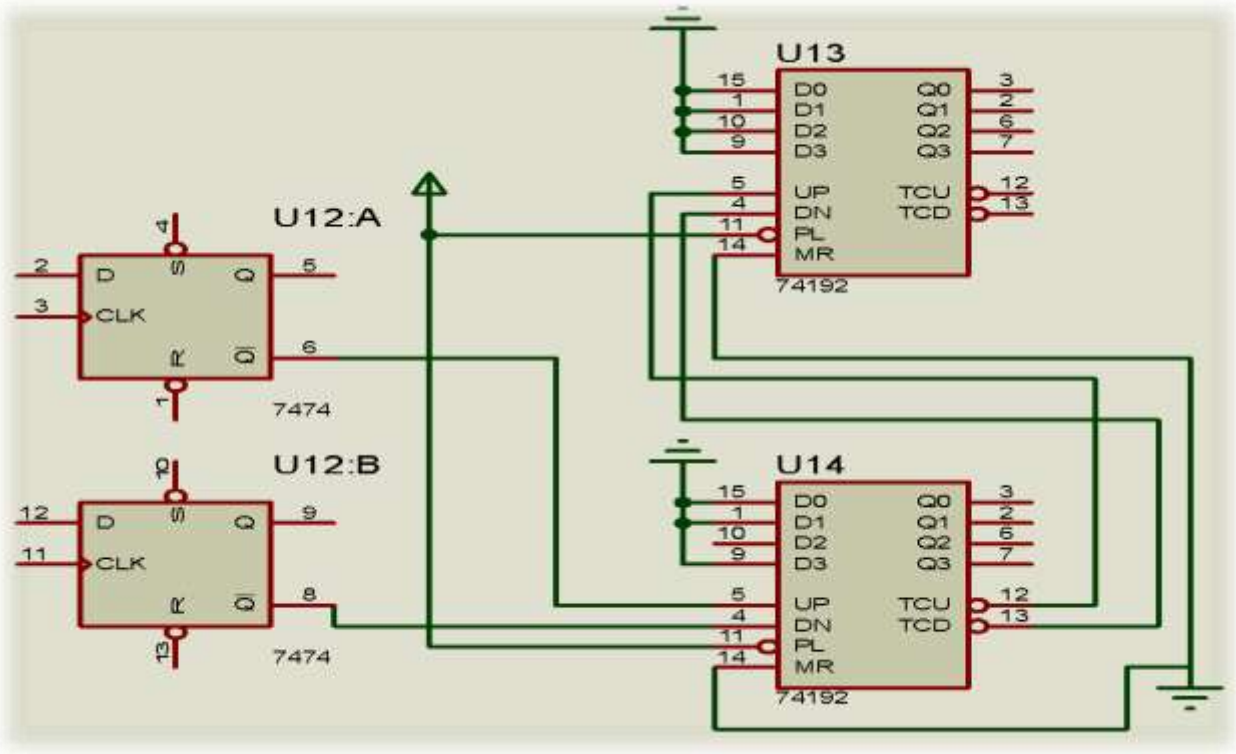


Figure 15:Counting Circuit

The carry-out (pin 12) and borrow-out (pin 13) of the first 74192 is connected to the clock input up (pin 5) and the clock input low (pin 4) of the second 74192, enabling proper tens digit counting.

4.3.4 Driver Circuit

The Driver Circuit utilizes the 7447 BCD-to-7-segment decoder/driver IC to convert the BCD output from the 74192 into a format suitable for controlling the 7-segment display. The BCD inputs (pins 7, 1, 2, and 6) of the 7447 are directly connected to the corresponding outputs of the 74192. The 7447 decodes the received BCD input and drives the 7-segment display through its output pins (13, 12, 11, 10, 9, 15, and 14), each of which corresponds to a specific segment (a-g) of the display. By simplifying the interfacing between the counter and the display, the 7447 ensures that the count is accurately represented on the 7-segment display.

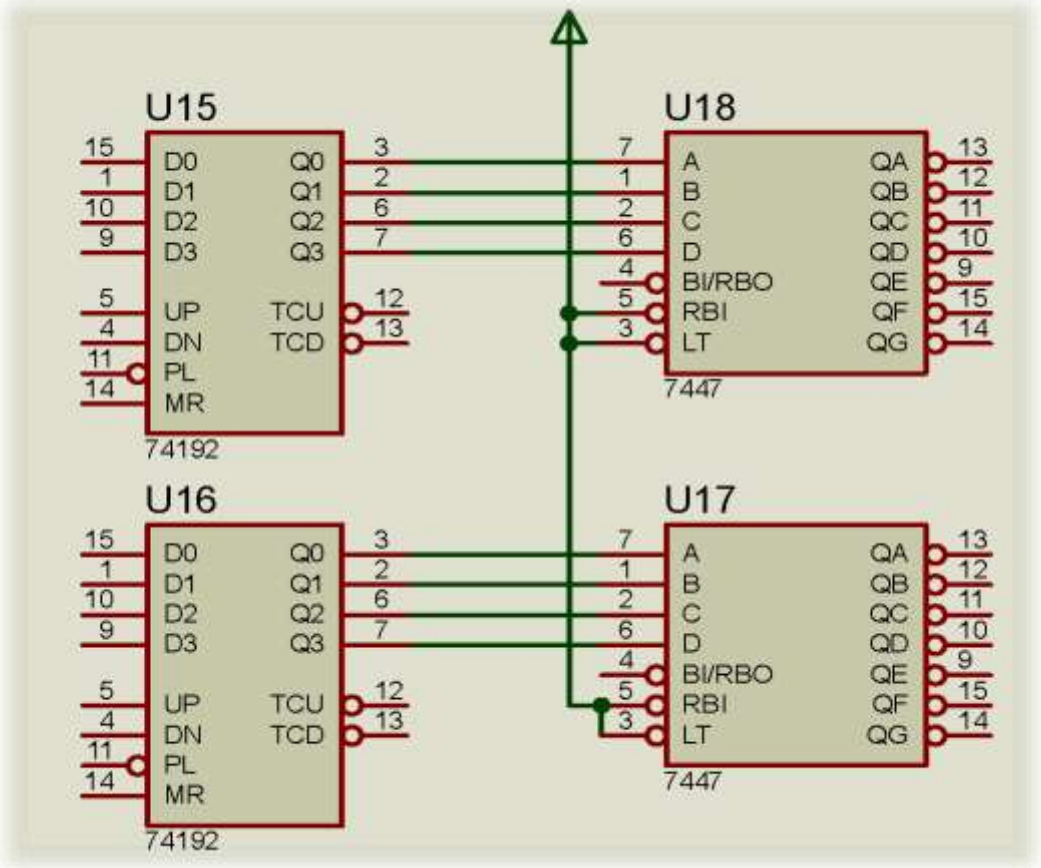


Figure 16:Driver Circuit

4.3.5 Displaying Circuit

The Displaying Circuit consists of two common cathode 7-segment displays to visually present the count. The 7447 IC manages the segment control, while the common cathode pins are connected to ground through current-limiting resistors. The first 7-segment display is responsible for displaying the units digit, while the second display shows the tens digit. To facilitate a 2-digit count, two 74192 counters and two 7447 decoder/driver ICs are used. The 7447 output pins (13, 12, 11, 10, 9, 15, and 14) are connected to specific segment of the display, (a-g) respectively. The common cathode configuration guarantees bright and easily readable displays. Additionally, the current-limiting resistors protect the LEDs in the 7-segment display from excessive current, thereby ensuring their durability and prolonged operational life.

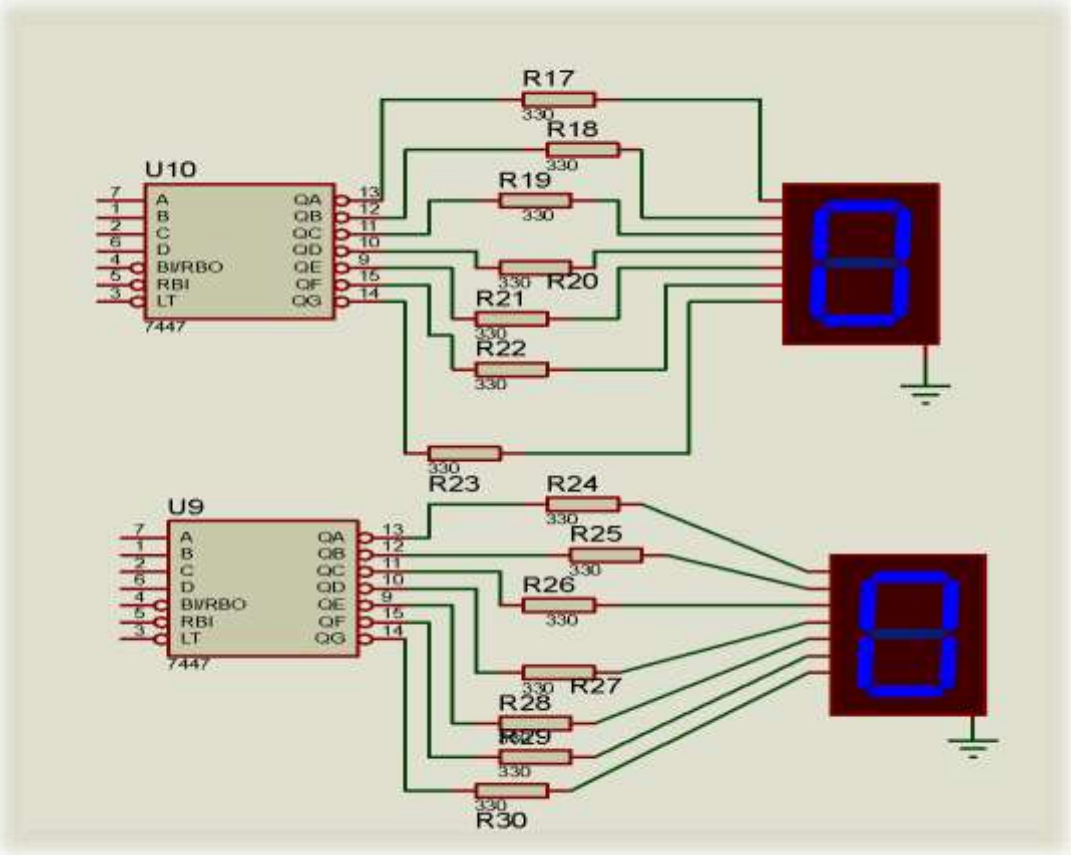


Figure 17:Displaying Circuit

4.4 Limitations

- When two or more objects are passed simultaneously at same speed, the object counter counts it as one.

4.5 Problems Encountered

- During testing, various issues such as unwanted multiple counts, unstable pulse generation, and logic mismatches were encountered and corrected.
- Initially, it was observed that the output of the LM358 was unstable due to minor light variations. To ensure smooth pulse generation, the reference voltage at the inverting input was carefully adjusted. Then, the LM358 provided a clear digital output for each object crossing.
- During practical analysis, initial miscounts occurred due to improper clock edge detection. Adjusting the timing characteristics of the Flip-Flop resolved the issue, ensuring that each object triggered only one stable pulse.

- During practical implementation, miscounts were observed due to glitches and short pulses. Ensuring a properly synchronized Flip-Flop output resolved this issue, making the counting process stable and accurate.
- Testing showed initial display errors due to incorrect wiring. Proper connections ensured that the 7447 correctly mapped the count to the 7-segment display without flickering or incorrect digit representation.

CHAPTER-5
CONCLUSION AND FUTURE WORKS

5.1 Conclusion

Object counting is one of the most important devices which we can use in different sectors in daily life. This is a bidirectional counting device which can count objects. This device would be portable and easy to use by the user. We want to bring this way of counting in real life and make people's lives easy and smart. We can use this system in factories to count the entering and leaving items in the factory and industry.

5.2 Future Works

- Test the system for longer periods to ensure consistent accuracy over time.
- Make the counting direction (up/down) adjustable by the user to provide more flexibility for different applications.
- Enclose all components in a more durable casing to protect the circuit and ensure the system operates reliably in different environments.
- Design a custom PCB with a double-layer copper board to improve circuit stability and reduce noise, ensuring more accurate counting.

REFERENCES

[1] Carvalho, B. F., de Melo Silva, C. C., Silva, A. M., Buiati, F., & de Sousa Júnior, R. T. (2016). Evaluation of an Arduino-based IoT Person Counter. In IoTBD (pp. 129-136).

[2] Sojol, J. I., Piya, N. F., Sadman, S., & Motahar, T. (2018). Smart bus: An automated passenger counting system. International Journal of Pure and Applied Mathematics, 118(18), 3169- 3177

[3] G. G. Roy and B. E. Wilkins (2019). *"Laser-based Visitor Counting Systems."* Journal of Applied Optics, 47(2), 110-117.

[4] Amin, S. M., Khan, M. T., & Riaz, Z. (2019). Visitor counting system using internet of things. *International Journal of Computer Applications*, 176(2), 1-7

[5] Alam, M., & Nayeem, M. (2017). "Design of Visitor Counting System using IR Sensors," *International Journal of Computer Applications*, 167(8), 23-26.

[6]. Kumar, A., & Rao, P. (2016). *RFID Technology for Visitor Counting and Access Control*. International Journal of Applied Engineering, 14(9), 124-132.

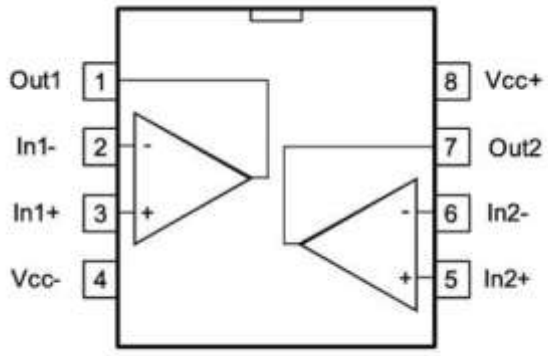
[7] S. Deepa, Y. R. Saravanakumar, P. Kirubakaran, K. Vijaykumar, M. Tharun. (2021). Bidirectional Visitor Counter with Automatic Room Light Controller, International Research Journal of Engineering and Technology (IRJET), 08(04), 479-481.

[8] A. S. F. Rahman, S. B. Yaakob, A. R. A. Razak, and R. A. Ramlee, "Post COVID-19 implementation of a bidirectional counter with reduced complexity for people counting application," J. Phys. Conf. Ser., vol. 1878, no. 1, 2021, doi: 10.1088/1742- 6596/1878/1/012040

[9] Matsui, K. (2005). *Electromechanical Visitor Counter Systems: Efficiency and Limitations*. Journal of Industrial Automation, 13(4), 32-40.

APPENDIX

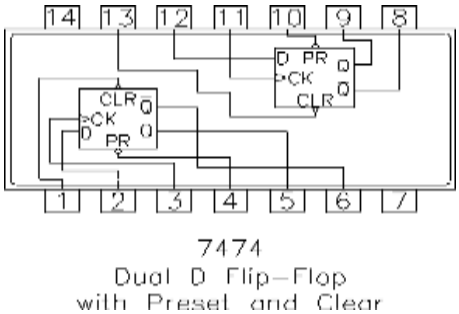
- Pin Configuration of IC LM358



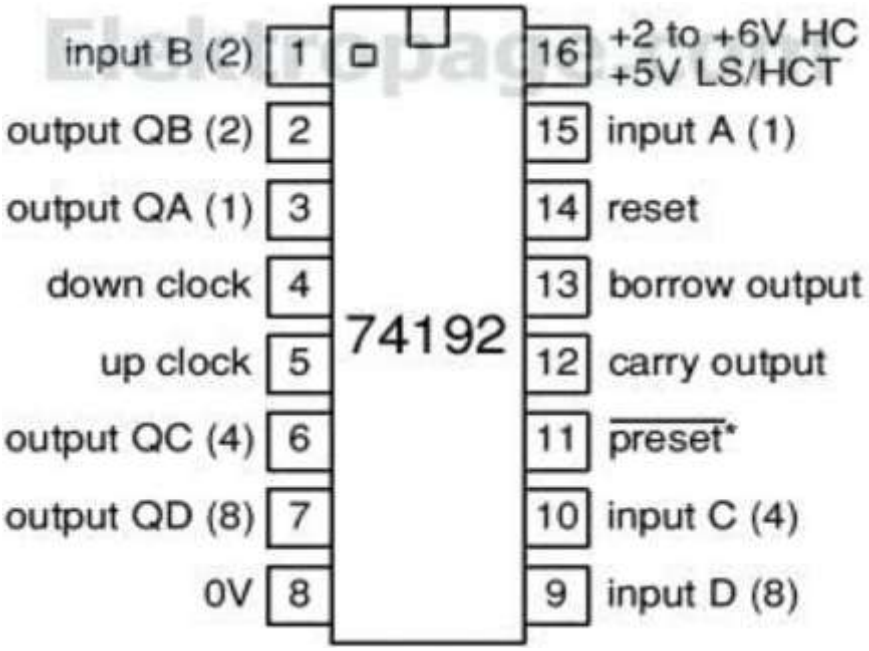
- Pin Description of IC LM358

pinout	name	descriptive
1	OUTPUT1	This pin is the output of first operational amplifier
2	INPUT1-	This pin is the inverting input of first op-amp
3	INPUT1+	This pin is the non-inverting input of first op-amp
4	VEE, GND	This pin is the non-inverting input of first op-amp
5	INPUT2+	This ground or negative supply to op-amp
6	INPUT2-	This pin is the non-inverting input of second op-amp
7	OUTPUT2	This pin is the output of the second op-amp
8	VCC	This pin is the positive voltage supply to both op-amp

- Pin Diagram of IC 7474



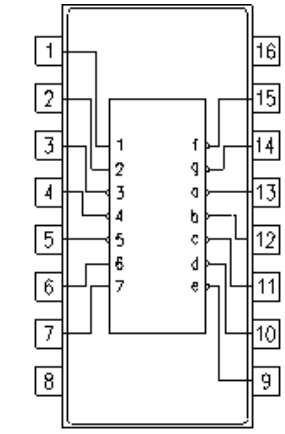
- Pin Configuration of IC 74192



- IC 74192 Truth Table

BCD count sequence				
Count	Output			
	D	C	B	A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H

- Pin Diagram of IC 7447



7447
BCD to 7-Segment
Decoder/Driver with (15V)
Open Collector Output

- Pin Description of IC 7447

Pin Number	Description
1	BCD B Input
2	BCD C Input
3	Lamp Test
4	RB Output
5	RB Input
6	BCD D Input
7	BCD A Input
8	Ground
9	7-Segment e Output
10	7-Segment d Output
11	7-Segment c Output
12	7-Segment b Output
13	7-Segment a Output
14	7-Segment g Output
15	7-Segment f Output
16	Vcc - Positive Supply